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***1. How many cars received the service in the simulation?***

In the simulation, a total of 42 cars were able to receive a car-wash service. This means that all these cars successfully went through the system and got cleaned. It shows how many customers were handled during the simulation process. This number helps us understand the capacity of the car-wash system.

***2. What time-related components were recorded in the simulation?***

The simulation recorded several important time-related components. These include the arrival time of each car, which shows when a car entered the system. It also recorded the waiting time, which tells how long a car stayed in line before getting serviced. Another is the service start time, which shows when the car wash actually began. Lastly, the service completion time shows when the car finished the process. These time details help in analyzing how efficient the system is.

***3. Give at least three (3) statistical computations that were performed in the problem.***

The simulation performed different statistical computations to evaluate the system. One example is the average waiting time, which shows how long cars usually wait before being serviced. Another is the average service time, which tells how long it typically takes to wash a car. It also computed the total number of cars served during the simulation. In addition, it included the utilization rate of the service facility, which shows how busy the system is. These calculations help us better understand the performance and efficiency of the car wash.

***4. How many random/pseudorandom number generators were used in the simulation?***

The simulation used two pseudorandom number generators. One generator was used to determine the arrival of cars, meaning it controls when cars enter the system. The other generator was used for the service time, which decides how long each car wash will take. These random values make the simulation more realistic because real-life situations are not always predictable.

***5. Can the simulation above be considered a stochastic simulation? Why or why not?***

Yes, the simulation can be considered a stochastic simulation because it involves randomness and uncertainty. The arrival of cars is not fixed, and the time it takes to wash each car can also change. Because of this, every time the simulation runs, the results may be different. This reflects real-life situations where outcomes are not always the same. Stochastic simulations are useful for modeling systems that depend on chance, like this car-wash service.

***6. Is the number of observations or simulation runs enough to deduce definitive inferences regarding the system? Why or why not?***

No, the number of observations or simulation runs may not be enough to make strong or final conclusions about the system. When there are only a few runs, the results can vary a lot and may not be consistent. This can lead to inaccurate or misleading conclusions. By increasing the number of simulation runs, we can get more stable and reliable results. More data helps us better understand the true performance of the system.

***7. In your opinion, what specific output analysis method is appropriate for the car-wash service simulation? Rationalize your answer.***

A good output analysis method for this simulation is averaging or statistical summary analysis. This method focuses on calculating averages like waiting time and service time. By using averages, we can easily understand the typical performance of the system. It also helps in comparing different simulation results. This method is simple to use and effective in showing how well the system is working overall. That is why it is suitable for this type of simulation.

***8. If you are to classify the car-wash service simulation, is it a terminating or a non-terminating simulation? Why?***

The car-wash service simulation is a terminating simulation because it has a clear starting point and ending point. For example, it may stop after serving a certain number of cars or after a specific amount of time has passed. Once that condition is reached, the simulation ends. This means it does not run continuously. Because it only focuses on a limited time or number of events, it is classified as a terminating simulation.

***9. Is it possible to create a frequency plot for the statistical report of the car-wash service simulation? Why or why not?***

Yes, it is possible to create a frequency plot using the data from the simulation. The simulation provides numerical values such as waiting times and service times. These values can be grouped into ranges and counted. A frequency plot shows how often certain values occur in the data. This makes it easier to see patterns and trends. It also helps in understanding how the system behaves over time. Because of this, frequency plots are very useful for analyzing simulation results.